ABDELRAHMAN MOHAMED AHMED

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PROFESSIONAL SUMMARY

Fresh Graduate Software Engineer and DevOps Engineer who builds production-style, cloud-native systems end to end — from Java/Spring Boot and Python/FastAPI microservices to React/TypeScript frontends. Comfortable across event-commerce, fintech, e-commerce, logistics, and AI project-planning domains, with hands-on exposure to distributed-systems patterns (Kafka, sagas, outbox), containerization and orchestration (Docker, Kubernetes, Terraform), CI/CD automation, and observability tooling (Prometheus, Grafana, OpenTelemetry). Also grounded in networking/cybersecurity, relational database design, signal processing, and machine learning fundamentals through coursework. Passionate about building meaningful systems, solving real-world problems, and continuously growing technical and communication skills.

EDUCATION

Modern Academy for Engineering and Technology
Bachelor of Computer Engineering — Graduated 2026

TECHNICAL PROJECTS

Event Management Platform

- Built a full event, ticketing, booking, payment, and notification platform as a set of Java/Spring Boot microservices (API Gateway, Auth, Event, Venue, Attendee, Payment, Notification) with a React/TypeScript frontend.
- Applied distributed-systems patterns — the outbox pattern, Kafka-driven notifications, and a payment/refund saga flow — secured end to end with JWT-based authentication.
- Instrumented every service for full observability with Prometheus, Grafana, Loki, OpenTelemetry, and Jaeger, and containerized the whole platform with Docker Compose.
- Delivered durable Kafka notification intents with templated email/SMS adapters, user preferences, and retry/dead-letter handling for resilient asynchronous messaging.

AI Project Manager Agent

- Designed an AI-powered planning platform that turns raw project ideas into structured plans, milestones, tasks, schedules, dependency graphs, risk views, and reports.
- Engineered a FastAPI/Python backend with database-backed background workers featuring job leasing, heartbeats, retries, and cancellation, plus provider-neutral AI adapters with validation ladders and repair loops for reliable generation.
- Built a React/TypeScript/Vite frontend with planning visualizations, graph and version APIs, human review-and-approval workflows, collaboration, and notifications.
- Hardened the system for real-world use with parallel planning workers, truncation-safe generation, observable planning runs, and a packaged Windows one-click launcher.

TWSELA — Courier & Delivery Management Platform

- Contributed to an enterprise courier and delivery platform for MENA logistics with Arabic-first RTL support across five roles: Owner, Admin, Merchant, Courier, and Warehouse.
- Worked across a Spring Boot 3.3.3 / Java 17 backend (Spring Security, JWT, JPA, Redis, WebSockets, Resilience4j) covering shipments, manifests, courier assignment, proof of delivery, and COD reconciliation.
- Built features on a React 18/Vite/Tailwind frontend, including real-time updates and analytics dashboards with Recharts.
- Authored architecture and audit documentation — security fixes, database analysis, and roadmap planning — to support production hardening.

Aircraft Management, Operation & Maintenance Using AI

- Contributed DevOps and infrastructure engineering to a large-scale aviation MRO platform covering maintenance compliance, forecasting, workforce planning, and AI-assisted operations.
- Built and maintained GitHub Actions CI/CD pipelines with Docker image builds, GHCR publishing, and Trivy/tfsec security scanning across a Java/TypeScript/Python stack.
- Managed Kubernetes deployments and AWS/EKS infrastructure with Terraform and Kustomize, resolving crash-loop errors, database privilege issues, and ingress configuration problems.
- Strengthened quality assurance through Playwright/Vitest test auditing, health checks, and frontend startup fixes.

CN Banking Transaction System

- Built a cloud-native banking transaction system in TypeScript covering account management, transfers, ledger, fraud detection, and notifications across independent microservices.
- Designed a production-style infrastructure stack — Docker Compose, Kubernetes, Terraform, and GitHub Actions CI/CD — with shared Kafka and types packages across services.
- Implemented transfer workflows, ledger logic, and fraud detection features, backed by observability tooling and platform health monitoring.
- Delivered a frontend portal with static export, authentication, routing, and isolated QA/testing.

MS Cloud-Native Platform

- Developed a cloud-native e-commerce microservices platform spanning a React/Vite frontend, an Nginx API gateway, and independent user (Node.js/Express), product (Python/FastAPI), and order (Node.js/Express) services.
- Implemented Redis caching and PostgreSQL-backed persistence with inter-service communication across the platform.
- Deployed the system to Kubernetes with horizontal pod autoscaling and blue-green deployments, provisioning AWS/EKS infrastructure with Terraform.
- Set up CI/CD pipelines with automated Docker image publishing and monitored the platform with Prometheus and Grafana.

StockPilot — Inventory & Sales Operations Platform

- Contributed to an inventory, sales, and fulfillment operations platform with a NestJS backend, a separate frontend application, and a dedicated Python ML service.
- Built Dockerfiles and multi-service Docker Compose environments for development and production across backend, frontend, and ML components.
- Set up GitHub Actions CI/CD pipelines with build gates and health checks per service, supporting staging and production deployment flows.

ACADEMIC PROJECTS

- **Insurance Company Software System** — Developed a desktop-style system for policyholders to view policies, track agent agreements, and access full policy history.
- **Database for Insurance Company System** — Designed a relational database aligned with the software system, with ERD modeling, normalization, and data-consistency rules.
- **Pulse Code Modulation (PCM) Simulation (MATLAB/Simulink)** — Built a full PCM workflow showing sampling, quantization, and encoding of analog signals into digital form.
- **Cybersecurity Hotel Network Project** — Designed a simplified hotel network and implemented firewalls, segmentation, and access control following core cybersecurity principles.
- **Machine Learning Flower Classification** — Built a classification model covering dataset preparation, feature extraction, training, and evaluation.
- **DevOps QR Code Scanner Project** — Built a Docker-based CI/CD pipeline that validated QR code detection, continuing the pipeline on success and halting safely on failure.

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript/JavaScript, C, MATLAB
- **Backend & Frameworks:** Spring Boot, Spring Security, FastAPI, NestJS, Node.js/Express, JPA
- **Frontend:** React, Vite, TypeScript, Tailwind CSS, Framer Motion, Recharts
- **Data & Messaging:** PostgreSQL, MySQL, Redis, Apache Kafka, Flyway
- **Cloud & DevOps:** Docker, Docker Compose, Kubernetes, Terraform, GitHub Actions, AWS/EKS, Kustomize, Trivy/tfsec
- **Observability:** Prometheus, Grafana, Loki, OpenTelemetry, Jaeger
- **Testing:** Playwright, Vitest
- **Security & Networking:** JWT, network segmentation, firewalls, access control fundamentals
- **Databases & Modeling:** ERD design, normalization, relational schema design
- **ML & Signal Processing:** Applied machine learning (training/evaluation), MATLAB/Simulink signal simulation

SOFT SKILLS

Public speaking • Communication • Cross-team collaboration on shared codebases